Love of Variety: Switching Stats

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1 Overview

My main update is around switching behavior of consumers. I have been mostly concerned with two things: length of switch and type of switch. To spoil, households will, on average, switch for about 1-1.5 periods. Further, most of the switching is from plain yogurt to some other flavor. I have left it as 12 flavors, as I worry I will be too zoomed out going to the proposed "berry, plain, other" setup, but if this is a major point, it would not be a problem to implement. Below, I will provide stats and a couple of graphs, as well as some definitions so the graphs are more interpretable.

2 Definitions and Statistics

Flavors are as follows:

1. Apple
2. Blueberry
3. Banana
4. Cherry
5. Key Lime
6. Lemon
7. Peach
8. Raspberry
9. Strawberry
10. Vanilla
11. Mixed Flavors
12. Plain

Table 1: Descriptive Statistics

Statistic	Value
# Households	7169
Mean # Times Observed	8.77
Yogurt Purchases	1.98
Mean Income	$17,164
Median Income	$17,000
% White	85%
% Black	11%
% Hispanic	3%
% Asian	2%
Variance in Flavor	0.54
Mean Repeat Purchase Rate (Brand)	69%
Mean Repeat Purchase Rate (Size)	92%
Mean Repeat Purchase Rate (Flavor)	50%
W/in HH Brand Concentration	42%
W/in HH Flavor Concentration	36%
Mean Consecutive Buys per HH	4.93
Mean % Return to Past Flavor After Switch	0.99

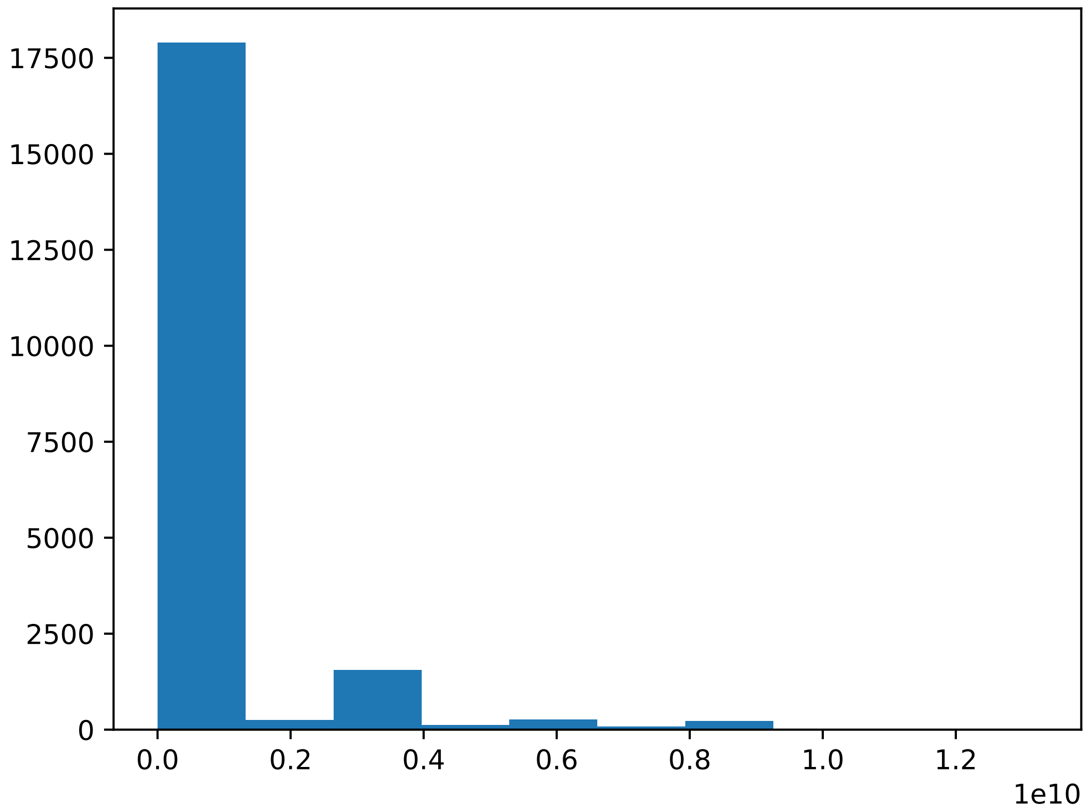
Statistics given before, updated from coding changes:

?? Shows that about half of the time, on average, households are buying the same flavor. Further, a singular flavor only makes up 36% of consumption. However, brand is more constant, implying that households may buy the same flavor within brand rather than swapping along that dimension to the same degree.

The most popular flavor is plain, and sees the most average consecutive buys. This statistic warrants further investigation, as it is purchased 13 consecutive shopping trips while the other flavors range between 1-5 consecutive trips. I will continue to look into this, but is at least a good benchmark.

You can see that on average, a household will buy the same flavor for about 5 periods, and 99% return to the prior flavor on average.

3 Figures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5000
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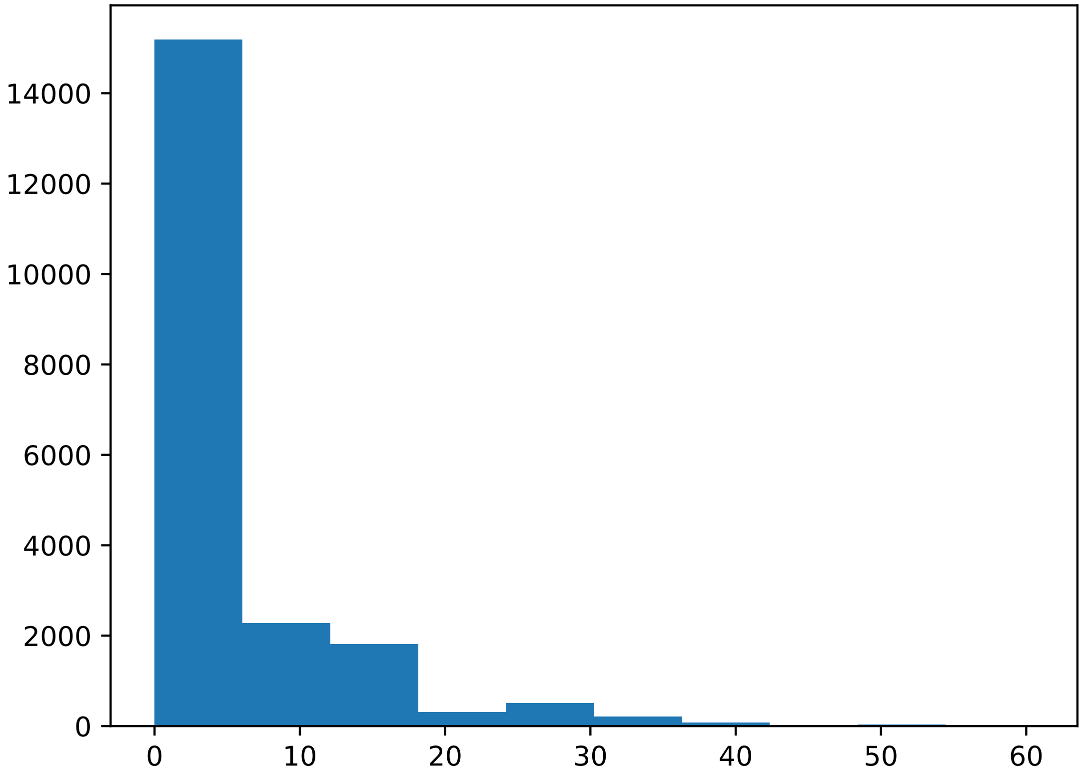
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40
50
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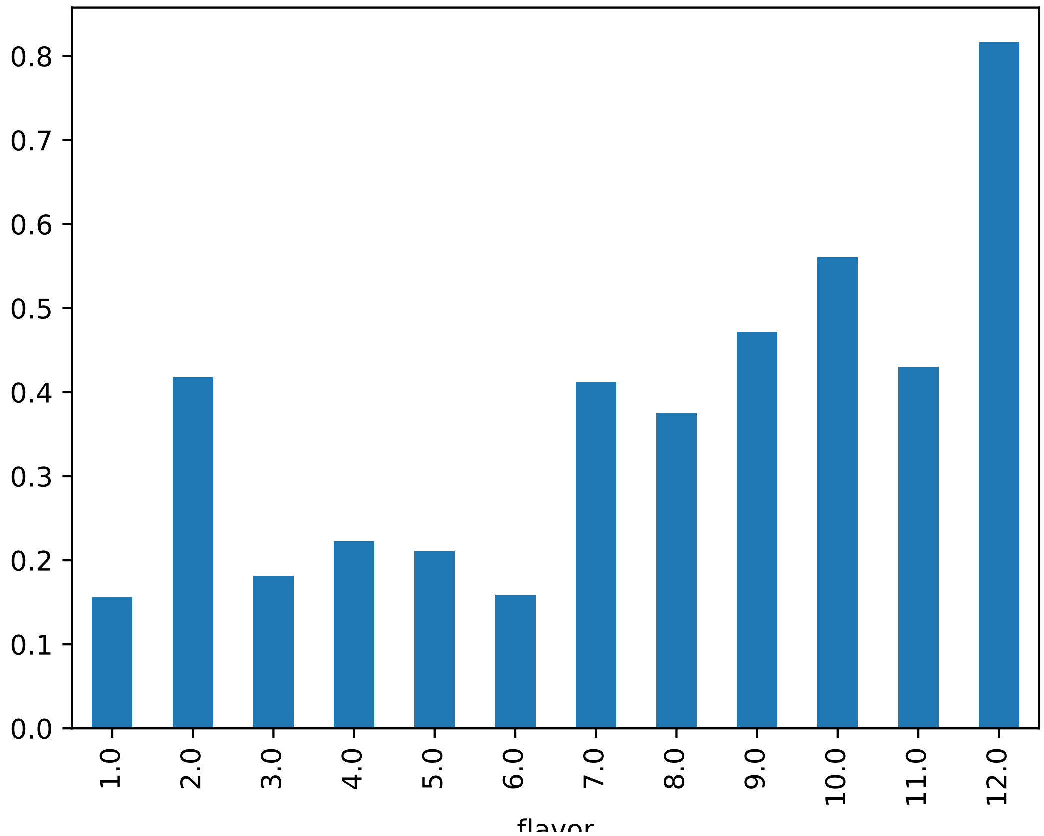
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0.7
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1.0
2.0
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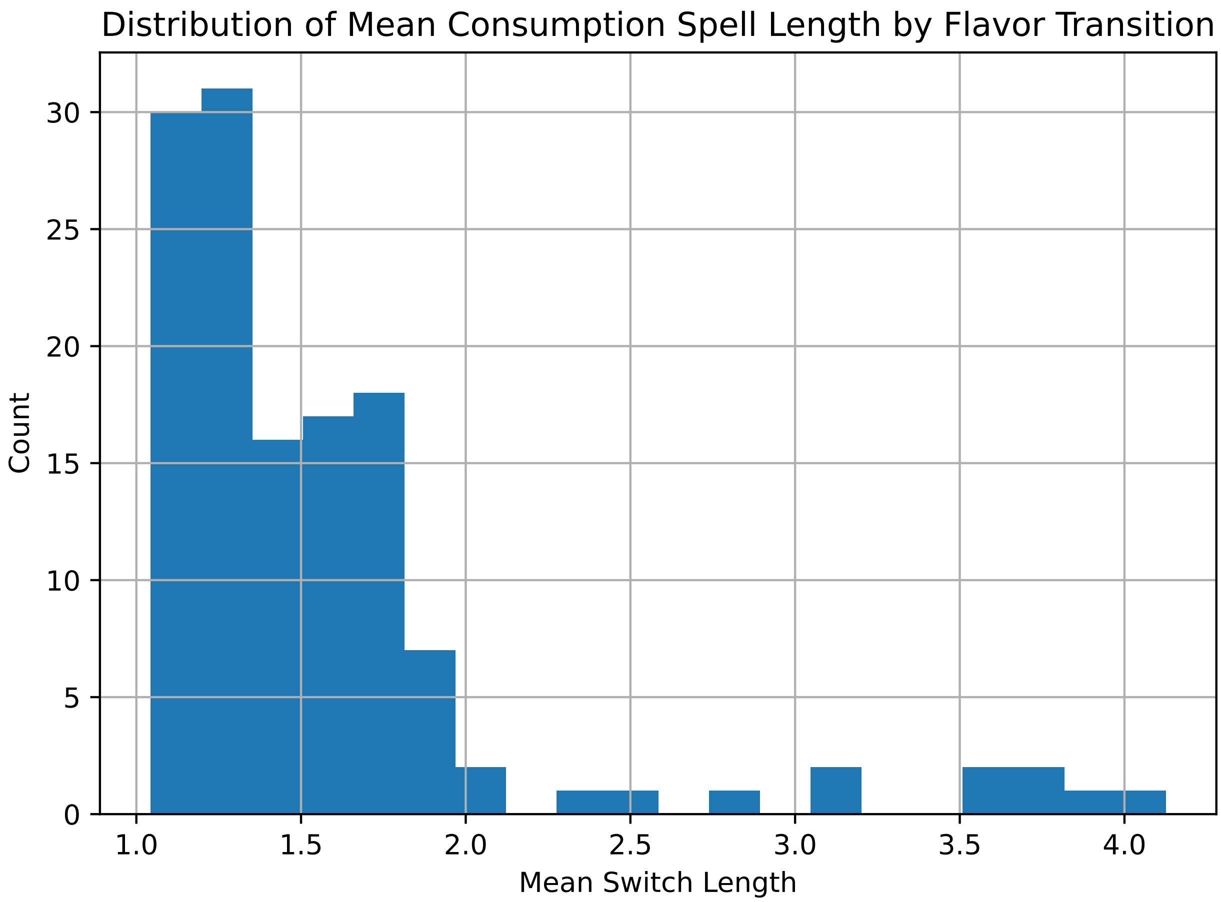
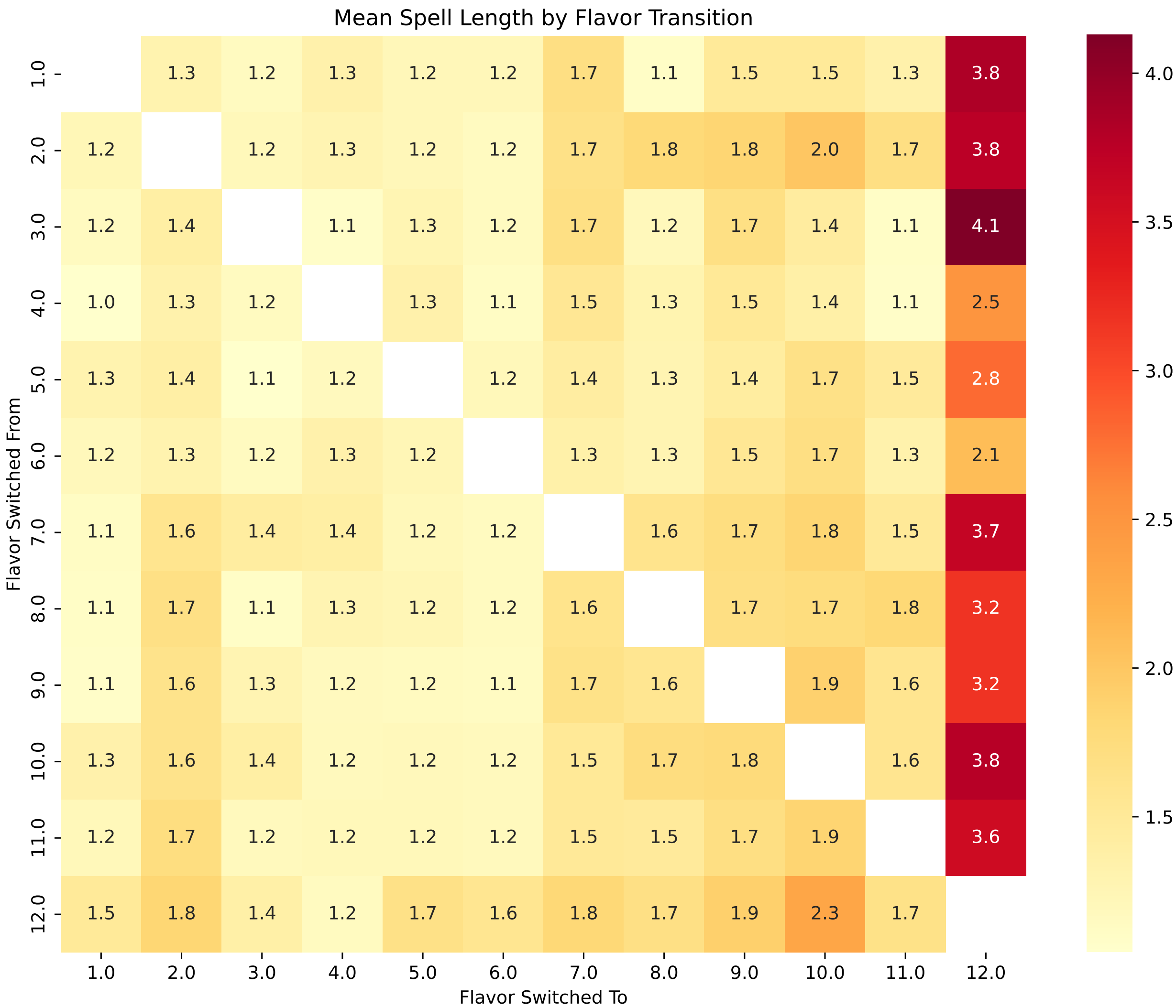
4.0
5.0
6.0
7.0
8.0
9.0
10.0
11.0
12.0
flavor



Distribution of Mean Consumption Spell Length by Flavor Transition
Count
Mean Switch Length
0
5
10
15
20
25
30
1.0
1.5
2.0
2.5
3.0
3.5
4.0



The two final graphs are new. With the first showing the distribution of how long people stay switched, with most households between one and one and a half periods. The heatmap shows the transitions. Most households are switching to plain, and staying on a flavor for 1-2 periods. Strawberry (10) is the most popular switch, with households staying between 1.5-2 periods.

4 Concluding and Next Steps

Any input would be great. Next, I want to divide by HH demographics to see if switching differs by things like income or race. I also think that continuing to formalize the transition matrix of flavors is important as it feels like it will be a useful statistic when we get to estimation.